

PRD — School Transport Operations System (TOS) MVP

MNC-grade Product Requirements + Technical Specification

Status: Finalized MVP scope (per “NO MORE CHANGES” decisions)

Audience: Product, Engineering, QA, DevOps, Security, Interns/Implementers

Integration Context: Embedded into existing ERP (fees are read-only from ERP)

This document is structured for direct implementation, QA sign-off, and operational readiness.

1) Executive Summary

This PRD defines an MVP for a School Transport Operations System (TOS) that enables:

- Admins to configure routes, assign students and drivers, and monitor operations.
- Drivers to start/end trips, stream GPS, and mark attendance per trip.
- Parents to receive a WhatsApp alert at trip start, then login to the ERP portal to view live bus location.
- System to enforce operational safety: idempotent notifications, no default attendance, driver cannot edit once marked, parent tracking requires login.

This MVP is intentionally designed to avoid over-engineering (no pre-created trips, no ETAs, no public links, no route ordering) while ensuring correctness, security, and a stable base for future scaling.

2) Problem Statement

Current operational pain:

- Parents lack timely updates on when pickup/drop starts.
- No real-time visibility into vehicle location (leading to calls, confusion, distrust).
- Attendance is often manual, inconsistent, and hard to audit.
- Transport teams lack centralized monitoring (driver activity, trip logs, attendance accuracy).

Why now:

Schools increasingly expect real-time transparency, auditability, and parent-facing tracking. The ERP needs a transport module that behaves like a simplified fleet system—reliable, secure, and operationally minimal.

3) Goals, Non-Goals, and Product Principles

3.1 Goals (MVP)

- Trip created at runtime when driver presses START (no pre-planned trips).
- WhatsApp notifications sent once per trip start (idempotent).
- Live location tracking available to parents via ERP after login.
- Attendance per trip, with no default state and driver edits blocked after marking.
- Admin visibility and correction capability with audit logs.
- Integrate transport fee as read-only from ERP.

3.2 Non-Goals (Strict “DO NOT BUILD” list for MVP)

- No ETA, no arrival prediction.
- No repeated notifications (only Trip Start).
- No public tracking link without login.
- No trip pre-scheduling or calendar-based trip generation.
- No student pickup ordering, optimization, or routing engine.
- No payment collection, fee editing, or transport fee rules management.
- No geofencing, speed alerts, driver scoring, fuel analytics (future).
- No offline-first complex sync (beyond minimal resilience).

3.3 Product Principles

- Operational correctness > feature richness
- Idempotency by default (notifications, trip start)
- Security-first parent experience (mandatory login)
- Auditable actions (admin edits and critical events)
- Low cognitive load for drivers (fast flows)

4) Stakeholders & Users

Personas:

1. Transport Admin (ERP Admin / School Ops)

- Creates routes, assigns students/drivers
- Monitors trips, live locations, attendance
- Corrects attendance with audit

2. Driver

- Logs in via mobile number
- Sees only assigned routes
- Starts trip (pickup/drop), sends GPS, marks attendance, ends trip
- Can view past attendance
- Can call parents

3. Parent

- Receives WhatsApp trip start message
- Must login to ERP portal to see tracking
- Views live bus location in Transport tab

5) MVP System Loop (Authoritative)

Driver starts trip → System creates trip → System sends WhatsApp notification batch (once) → Driver app streams GPS every 10-15 seconds (background) → Parent logs in → Transport tab → views live location → Driver marks attendance per student (present/absent) → Driver ends trip → System stores trip summary + logs

6) Functional Requirements (Detailed)

6.1 Authentication & Access Control

Roles:

- ADMIN
- DRIVER
- PARENT

Requirements:

- Driver login via mobile number (OTP recommended).
- Parents must login (ERP auth) to view tracking.
- Drivers can access only assigned routes.
- Admin can access routes, trips, attendance, live map for the school(s) they manage.

Acceptance Criteria:

- Parent cannot see tracking without a valid session.
- Driver cannot start a trip for an unassigned route.

6.2 Route System

Definition:

- Route = a list of students (unordered in MVP).
- Route has a driver assignment (one primary driver for MVP).

Admin Capabilities:

- Create route
- Assign/remove students to route
- Assign driver to route

Constraints (MVP):

- No ordering of students.
- One driver per route at a time (extendable later).

Acceptance Criteria:

- Route can exist without students (config stage).
- Route with students can be assigned to a driver.
- Driver sees only their routes.

6.3 Trip System (Runtime Trip Creation)

Definition: Trip is created only when driver clicks START.

Trip must include:

- route_id
- driver_id
- trip_type = PICKUP or DROP
- trip_date (derived)
- start_time, optional end_time
- status = ACTIVE/ENDED

Rules:

- No pre-created trips.
- Trip is tied to driver + route + type + date.
- System should prevent duplicates: only one ACTIVE trip per route + trip_type at a time (per tenant/school).

Driver Actions:

- Select route
- Select Pickup/Drop
- Start Trip
- End Trip

Acceptance Criteria:

- Trip record is created at start.
- Trip status becomes ACTIVE and blocks another trip of same type for same route (unless ended).
- End Trip closes the trip and stops active tracking status.

6.4 GPS Tracking System (Driver Mobile → Backend)

Source: Driver mobile app

Frequency (MVP):

- Every 10–15 seconds
- Background mode supported

Storage Requirements:

- Latest location table/store (for live map)
- Historical logs table/store (for analytics later)

Data captured per GPS ping:

- trip_id (must be associated to active trip)
- driver_id
- route_id
- timestamp
- lat, lng
- accuracy_m, speed (optional but recommended)
- battery (optional future)
- device_time (optional)
- server_received_time

Acceptance Criteria:

- Parents' map view updates with latest location within acceptable delay (e.g., <30s typical).
- Historical logs recorded for each ping during trip.

6.5 Notification System (WhatsApp — Trip Start only)

Trigger: Trip Start event

Audience:

- For MVP: all students on route receive a trip-start message for their parent(s).
- Trip Start is not dependent on attendance.

Message Content (MVP):

- Student Name
- Pickup started or Drop started
- Link to ERP portal (requires login)

No ETA. No repeated notifications.

Idempotency Rule (Critical): One trip → one notification batch only.

Implementation requirements:

- notification_log enforces uniqueness on (trip_id, notification_type='TRIP_START')
- If retry occurs, system returns success without re-sending.

Acceptance Criteria:

- If driver taps Start twice or network retries occur: only one WhatsApp batch is sent.
- Notifications are logged with status (queued/sent/failed).

6.6 Attendance System (Per Trip)

Definitions:

- Attendance is recorded per trip (pickup and drop separately).
- Values: Present, Absent.
- No default state (must be explicitly marked).
- Driver cannot edit after marking.
- Admin can edit with audit log.

Driver Flow:

- During active trip, driver views student list for the route.
- Driver marks each student as Present or Absent.
- Once marked, that student's status is locked for the driver.

Admin Flow:

- Admin can edit attendance entries.
- Every admin edit must create an audit log entry: who edited, when, old value, new value, reason (recommended mandatory).

Acceptance Criteria:

- Unmarked student remains Not marked (null) until explicitly set.
- Driver cannot modify after save.
- Admin edits visible with audit history.

6.7 Parent Portal (Mandatory Login)

Entry Point: WhatsApp message includes link (e.g., <https://erp.school.com/transport>).

Flow:

WhatsApp → Click link → Login (if not logged in) → Transport Tab → Live Bus Map

Live Map Requirements:

- Display route/bus label, last updated timestamp.
- Show bus marker based on latest location.
- Optional: show Trip active status.

Acceptance Criteria:

- Without login, parent is redirected to login.
- After login, parent sees tracking for their child's assigned route(s) only.

6.8 Admin Dashboard (Operations Console)

Admin can:

- Create routes
- Assign students to routes
- Assign drivers
- View live bus locations (active trips)
- View attendance per trip and per route
- View driver activity (trip start/end times, last GPS timestamp)
- View transport fee (read-only)

Live Monitoring Requirements:

- Active Trips list
- Last GPS ping time and health indicator: healthy (<30s), warning (30-90s), stale (>90s; configurable)

Acceptance Criteria:

- Admin can open an active trip and see location + student attendance progress.

6.9 Fee System (Read-only ERP Integration)

Requirement: Transport fee is fetched from existing ERP data. No editing in TOS module.

Acceptance Criteria:

- Fees visible in admin dashboard and/or parent transport tab (as decided by ERP UX).

- All fee modifications are out of scope.

7) Non-Functional Requirements (NFRs)

7.1 Security

- Mandatory authentication for parents (no public links).
- RBAC enforcement at API level (not only UI).
- Data isolation per school/tenant.
- Signed tokens (JWT) with short expiry; refresh flows as per ERP standards.
- WhatsApp content must not leak sensitive location without login requirement.

7.2 Reliability

- GPS ingestion tolerates intermittent network: client retries with backoff.
- Server idempotency for duplicate pings optional; can deduplicate by (trip_id, timestamp, lat, lng).

7.3 Performance

- Live location query should be fast (read from latest location store keyed by route/trip).
- GPS write throughput sized for peak: $\#buses * 6 \text{ pings/min (10s)}$ to $\#buses * 4 \text{ pings/min (15s)}$.

7.4 Observability

- Structured logs: trip start/end, notification attempts/results, GPS ingestion counts/last ping, attendance events.
- Metrics: notification success rate, GPS receive rate, active trips count, stale trackers count.

7.5 Auditability

- Admin attendance edits must generate immutable audit records.

8) System Architecture (MNC-Style)

8.1 High-level Components

- Driver Mobile App: Auth (mobile OTP), assigned routes, trip control, attendance, background GPS streaming.
- ERP Web Portal (Parent + Admin): Parent Transport tab with live map; Admin dashboard and monitoring.
- Backend Services (modular monolith initially): Auth integration, Route, Trip, GPS ingestion, Attendance, Notification, Reporting/queries.
- Data Stores: Relational DB; optional Redis for latest location.
- External Integrations: WhatsApp provider; ERP Fees API (read-only).

8.2 Reference Data Flow (Text Diagram)

Trip Start:

- Driver App → POST /trip/start
- Trip Service creates trip (transaction)
- Notification Service enqueues TripStart notifications
- Notification Log enforces idempotency

- WhatsApp Provider sends messages
- GPS:
- Driver App → POST /gps/update every 10–15s
- GPS Service writes: latest_location (overwrite) and gps_logs (append)
- Parent Tracking:
- Parent Portal → GET /tracking/live?route_id=...
- Tracking API returns latest location (and last updated)
- Attendance:
- Driver App → POST /attendance/mark
- Attendance locked for driver after first write
- Admin edits via PATCH /admin/attendance/{id} with audit log write

9) Data Model (Logical)

Core Tables (suggested):

- users (id, role, tenant_id, phone, status)
- students (id, tenant_id, name, parent_user_id, ...)
- routes (id, tenant_id, name, status)
- route_students (route_id, student_id)
- route_driver_assignment (route_id, driver_user_id, active_from, active_to)
- trips (id, tenant_id, route_id, driver_id, type, status, start_time, end_time, created_at)
- attendance (id, trip_id, student_id, status [PRESENT/ABSENT], marked_by, marked_at, locked=true)
- attendance_audit (id, attendance_id, edited_by_admin, old_status, new_status, reason, edited_at)
- latest_bus_location (trip_id, route_id, driver_id, lat, lng, accuracy, updated_at)
- gps_logs (id, trip_id, timestamp, lat, lng, accuracy, received_at)
- notification_log (id, trip_id, type, idempotency_key, status, provider_msg_id, created_at)

Critical Constraints:

- Unique index: notification_log(trip_id, type)
- Unique/guard: one ACTIVE trip per (route_id, type) (enforce via DB constraint or transactional check)

10) API Specification (Contract-Level)

Trip

- POST /trip/start — body: { route_id, trip_type } — returns: { trip_id, status, started_at }
- POST /trip/end — body: { trip_id } — returns: { trip_id, status, ended_at }
- GET /trip/active?route_id=...&trip_type=...

GPS

- POST /gps/update — body: { trip_id, lat, lng, accuracy_m, timestamp }
- GET /tracking/live?route_id=... — returns: { lat, lng, updated_at, trip_id, trip_type }

Attendance

- POST /attendance/mark — body: { trip_id, student_id, status } — rule: if already marked by driver → reject (409) or return locked state
- GET /attendance?trip_id=...

Notifications (internal)

- POST /notify/trip-start (invoked by backend on trip start event; uses idempotency constraint)

Admin

- POST /route/create
- POST /route/assign-driver
- POST /route/assign-students
- GET /admin/live-trips
- PATCH /admin/attendance/{attendance_id} (with audit reason)

11) Edge Cases & Handling (Mandatory)

Trip Start

- Double tap Start → second request returns existing trip or rejected; notifications not duplicated.
- No students on route → trip can start; notification batch is empty; log should reflect.
- Driver starts trip without GPS permission → allow trip but show warning and mark GPS health as stale.

GPS

- Network intermittent → app retries; backend accepts late pings (store received_at).
- GPS pings without active trip → reject with 400/409; app prompts driver to start trip.
- Multiple devices sending GPS for same driver (rare) → accept; last-write-wins for latest_location; log includes device id if available.

Attendance

- Student not marked by end of trip → allowed (no default). Admin dashboard highlights unmarked.
- Driver tries to change marked attendance → blocked.
- Admin edits → always logged with old/new value and reason.

Parent Tracking

- Parent clicks link but not logged in → redirect to login; after login, deep link to Transport tab.
- Parent has multiple children/routes → show list or default view with selectable child/route.

Notifications

- WhatsApp provider failure → retry queue (still idempotent). No duplicates; status updated in log.
- If retry later succeeds, mark as sent with provider message id.

12) Success Metrics (KPIs)

Product KPIs

- % of trips where parents successfully open tracking after WhatsApp click
- Reduction in transport-related calls to school (proxy via admin feedback)
- Driver adoption: trips started / expected trips per day

System KPIs

- Notification success rate (sent/attempted)
- Median time from trip start → notification delivered (provider-dependent)
- GPS freshness: % time location updated within 30s during active trips
- Attendance completeness rate: % students marked by end of trip

13) Epics, Features, User Stories (With Acceptance Criteria)

EPIC A — Foundations: Auth, RBAC, Tenant Isolation

- A1. Driver OTP login — AC: invalid OTP blocked; role verified; token issued.
- A2. Parent login enforced — AC: unauthenticated requests redirect/deny.
- A3. RBAC enforcement — AC: parent cannot access admin APIs; driver cannot access other routes.

EPIC B — Route Management

- B1. Create route — AC: route appears in admin list.
- B2. Assign students — AC: route-student mapping persists; driver sees updated list.
- B3. Assign driver — AC: driver sees route in app; other drivers don't.

EPIC C — Trip Runtime System

- C1. Start trip — AC: trip created; becomes ACTIVE; start_time stored.
- C2. Prevent duplicate active trips — AC: second request returns error or existing trip.
- C3. End trip — AC: trip status ENDED; end_time stored.

EPIC D — GPS Tracking

- D1. Background GPS streaming — AC: backend receives pings; latest location updates.
- D2. Latest location storage — AC: parent view reflects near-real-time location.
- D3. Historical logs — AC: logs exist for analytics/export later.

EPIC E — Notifications (WhatsApp)

- E1. Trip-start notification batch — AC: message includes student name + pickup/drop started + portal link.
- E2. Idempotent notification — AC: retries do not duplicate sends; uniqueness enforced.
- E3. Delivery status tracking — AC: admin can see success/failure counts (optional MVP+).

EPIC F — Attendance

- F1. Mark attendance — AC: no default; saved values persist.
- F2. Lock attendance from driver edits — AC: server rejects changes.
- F3. Admin edit with audit — AC: audit record generated with old/new values.

EPIC G — Parent Live Tracking

- G1. Transport tab UI — AC: map loads, marker updates, last-updated shown.
- G2. Multi-child support — AC: correct route location shown.

EPIC H — Admin Dashboard & Monitoring

- H1. Live buses view — AC: stale indicator visible.
- H2. Attendance monitoring — AC: counts present/absent/unmarked.
- H3. Driver activity log — AC: trip list filter by date/route/driver.

EPIC I — ERP Integration (Fees Read-only)

- I1. Fee display — AC: read-only; errors handled gracefully.

14) Testing Strategy (MNC-grade)

14.1 Test Levels

Unit Tests

- Trip creation rules; one active trip per route/type; notification idempotency enforcement.
- Attendance locking + admin edit audit; permission checks (RBAC).

Integration Tests

- WhatsApp provider sandbox flow (mock/stub).
- ERP fee API integration (mock/stub).
- GPS ingestion pipeline (latest + logs).
- Parent tracking endpoint returns latest location.

End-to-End (E2E)

- Driver: login → start trip → send GPS → mark attendance → end trip.
- Parent: receive message → login → view live tracking.
- Admin: monitor live trip → edit attendance with audit.

14.2 Special Test Scenarios

- Duplicate trip start requests (same driver, same route).
- Duplicate notification requests (retries).
- GPS burst traffic (load test).
- Stale GPS detection.
- Attendance left unmarked and admin completion later.

14.3 QA Acceptance Gate (Release Criteria)

- Zero duplicate notification cases in test suite.
- 95%+ WhatsApp send success in staging tests (provider-dependent).
- Parent cannot view location without login (verified).
- Attendance audit log created on every admin change (verified).

15) Deployment Plan (ERP-Integrated)

Environments

- Dev: rapid iteration.
- Staging: provider sandbox + ERP sandbox.
- Production: controlled rollout by school/tenant.

Rollout Strategy

- Feature flags: enable transport module per tenant; enable WhatsApp notifications per tenant.
- Gradual rollout: pilot 1-2 schools → expand.

Operational Runbooks

- WhatsApp provider outage handling (queue + retries).
- GPS ingestion spikes (autoscale if applicable).
- Incident response: tracking not updating → check active trip, last ping, driver device permissions, backend health.

16) Risks & Mitigations

- Driver app background GPS restrictions (Android/iOS) → foreground service on Android; clear permission UX; show GPS health.
- WhatsApp provider delivery variability → retry queue; status logging; idempotency prevents duplicates.
- Data consistency (attendance) → no default; server lock; admin edits only with audit.
- Security/privacy → mandatory login; RBAC; tenant isolation; avoid public location sharing.

17) Execution Plan (Suggested 5 Sprints)

- Sprint 0 (Setup): repo setup, CI/CD skeleton, DB migrations, core RBAC middleware.
- Sprint 1 — Routes + Auth: driver OTP login; admin route CRUD + assign students + assign drivers.
- Sprint 2 — Trip System: start/end trip; active trip constraints; admin trip views (basic).
- Sprint 3 — GPS + Parent Tracking: GPS ingestion + latest location store; parent transport tab live map (auth enforced).
- Sprint 4 — Attendance: driver attendance marking + locking; admin edit + audit log UI/API.
- Sprint 5 — WhatsApp Notifications + Hardening: WhatsApp integration; idempotency log + retries; observability + load tests; staging + pilot rollout.

18) Appendices

A) Glossary

- TOS: Transport Operations System
- Trip: Runtime entity created at driver start
- Idempotency: Ensuring duplicate requests do not cause duplicate outcomes
- Tenant: School or organization boundary in ERP

B) Minimum UI Screens (Deliverables Checklist)

Driver App

- Login
- Route list
- Trip start (route + pickup/drop)
- Live status + GPS indicator
- Attendance marking screen
- End trip
- Past attendance view

Parent Portal

- Login

- Transport tab with map + last updated + trip status

Admin Dashboard

- Route management (students/drivers assignment)
- Live trips map/list
- Attendance viewer + admin edit modal with reason + audit trail
- Driver activity list